Foundation A · Cell Biology

40 marks • About 45 minutes • Scientific calculator permitted.

Original Study Hub review paper for AQA Trilogy (8464), not an official AQA paper. Answer every part; show working. For “Choose one”, circle one letter. Answer spaces are indicative. Marking guides are separate.

1. Cells and their structures · 5 marks

1a [1 mark]

Which structure is normally present in plant, animal and bacterial cells? Choose one.

A. Cytoplasm

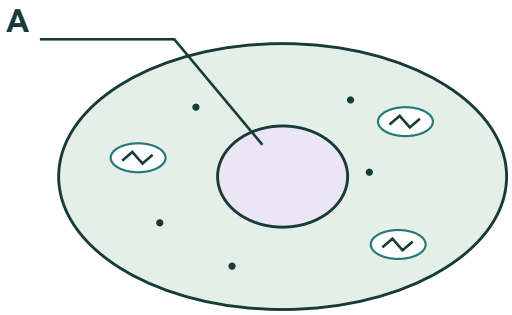
B. Nucleus

C. Chloroplast

1b [2 marks]

Figure F-CELL shows an animal cell. Name structure A. State its function.

F-CELL · Animal cell



Animal cell • schematic, not to scale

1c [1 mark]

Which description fits a bacterial plasmid? Choose one.

A. A site of aerobic respiration

B. A small ring of DNA

C. A membrane-bound nucleus

1d [1 mark]

An unknown cell has a cellulose wall and a large permanent vacuole, but no chloroplasts. Which identification best fits? Choose one.

A. Plant cell

B. Animal cell

C. Bacterial cell

2. Microscopy and RP01 · 6 marks

2a [1 mark]

A microscope has ×4, ×10 and ×40 objectives. Which should be used first to locate a specimen? Choose one.

A. ×10

B. ×40

C. ×4

2b [1 mark]

An electron microscope can show two very close structures separately. Name the property that makes this possible.

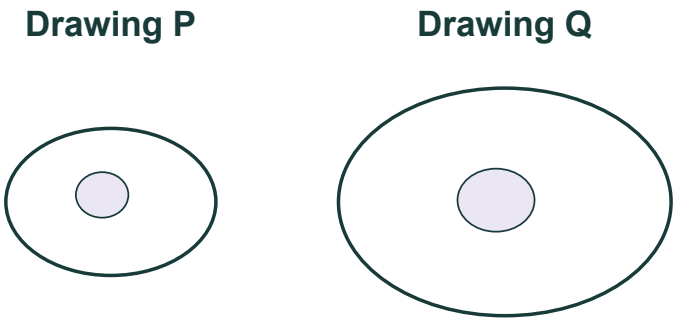
2c [3 marks]

The recorded image length is 24 mm and the magnification is ×600. Calculate the real cell length in micrometres. Use real length = image length ÷ magnification, and 1 mm = 1000 micrometres. Show working.

2d [1 mark]

The two observation drawings in F-DRAWING are from different specimens. Explain why their displayed lengths cannot be used to decide which cell was really larger.

F-DRAWING · Two observation records



Student observation drawings

3. Cell division and specialisation · 6 marks

3a [1 mark]

Which statement about a chromosome is correct? Choose one.

A. Each chromosome carries exactly one gene

B. A chromosome is made of DNA and carries many genes

C. A chromosome is made of genes but contains no DNA

3b [2 marks]

Before mitosis, a cell grows and increases its sub-cellular structures. Describe what happens to its DNA before mitosis and what happens to chromosomes/nuclei during mitosis.

3c [1 mark]

Which statement about differentiation is correct? Choose one.

A. All mature animal cells can become any cell type

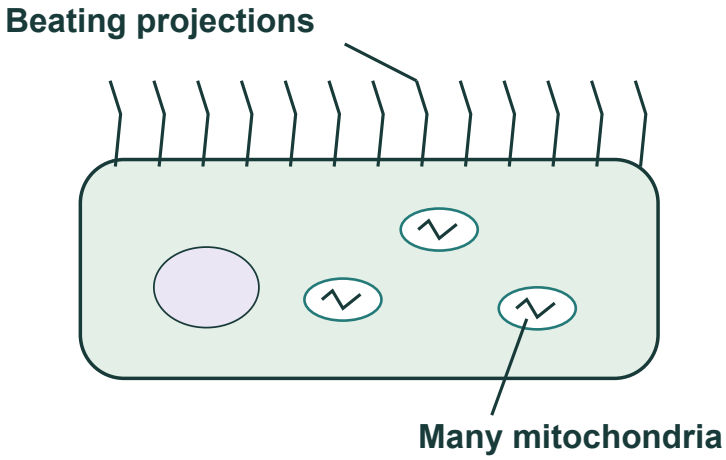
B. Differentiation means a cell divides into two identical cells

C. Many plant cells can differentiate throughout life

3d [2 marks]

F-CILIATED shows a cell from an unfamiliar water animal. Its hair-like projections beat to move liquid, and it contains many mitochondria. Explain how the mitochondria help it do this.

F-CILIATED · Unfamiliar cell



Schematic, not to scale

4. Stem cells and evidence · 5 marks

4a [2 marks]

State one cell type adult bone-marrow stem cells can form. State how embryonic stem cells differ in the range of cells they can form.

4b [1 mark]

Which description is therapeutic cloning? Choose one.

A. Mature blood cells are collected directly for transfusion

B. Adult bone-marrow cells are removed and transplanted without making an embryo

C. An embryo with the same genes as the patient is produced

4c [2 marks]

A research team proposes further work on the treatment in F-STEM-TABLE. Explain one reason to continue the research and one reason it may not be suitable for every patient.

F-STEM-TABLE · Experimental treatment

F-STEM-TABLE · Experimental treatment

Feature	Information
Potential outcome	Could replace damaged nerve cells
Safety observation	A virus could be transferred
Cell source	Embryos are used

5. Diffusion and exchange · 6 marks

5a [1 mark]

Which statement defines diffusion? Choose one.

A. Net particle movement from higher to lower concentration

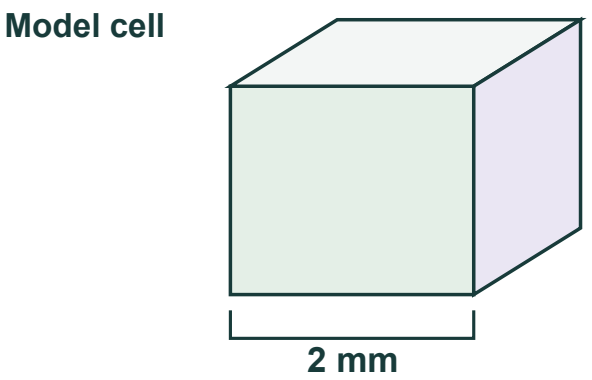
B. Net particle movement from lower to higher concentration using respiration energy

C. Only water moves through a partially permeable membrane

5b [2 marks]

A model cell is a cube with side length 2 mm. Calculate its surface-area-to-volume ratio. Use surface area = 6 × side² and volume = side³. Show working.

F-CUBE · Cube model



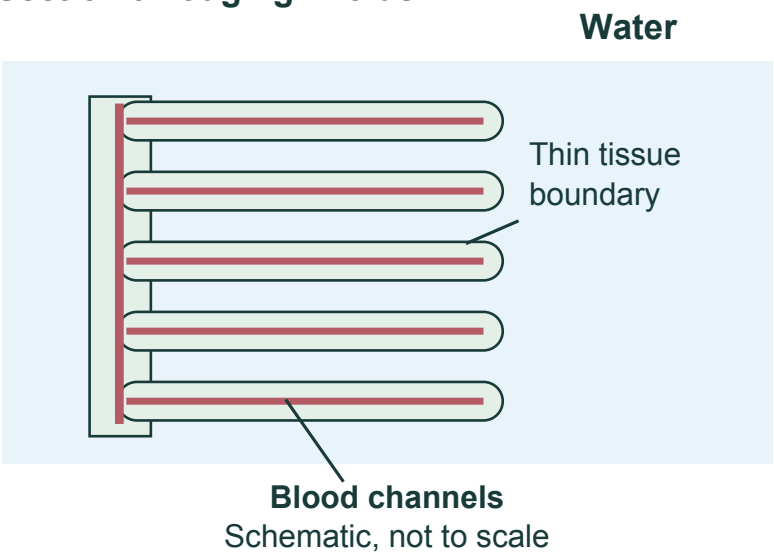
5. Diffusion and exchange · continued

5c [2 marks]

Figure F-GILL shows many thin folds with a good blood supply. Explain how two of these features increase gas exchange.

F-GILL · Gill folds

Section through gill folds



Schematic, not to scale

5d [1 mark]

The table gives mean diffusion rates of 0.30 units/min at 10 °C and 0.60 units/min at 20 °C, with all other conditions controlled. Which conclusion is supported? Choose one.

F-DIFF-TABLE · Controlled diffusion test

F-DIFF-TABLE · Controlled diffusion test

Temperature / °C	Mean rate / units/min
10	0.30
20	0.60

A. Diffusion was faster at 20 °C under these conditions

B. Every 10 °C rise must double diffusion at every temperature

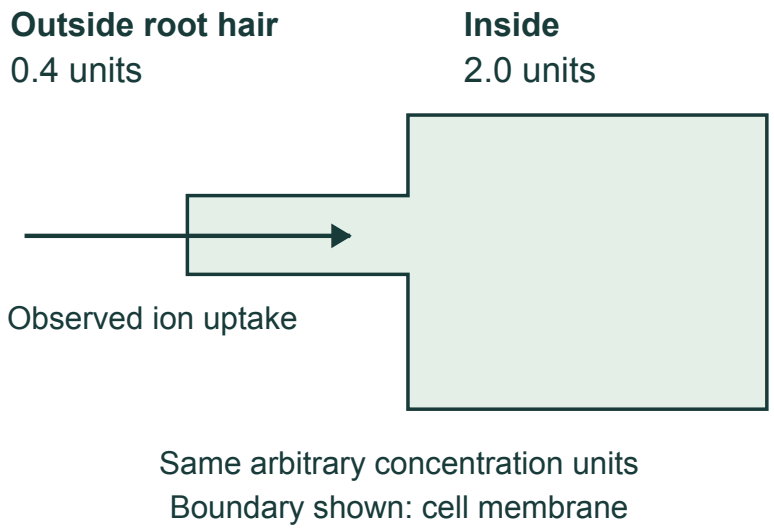
C. Temperature supplied the respiration energy needed for diffusion

6. Transport processes · 4 marks

6a [2 marks]

Figure F-ION shows mineral-ion concentration 0.4 units outside a root hair and 2.0 units inside, with uptake into the cell. Explain how uptake can continue.

F-ION · Mineral-ion uptake



6b [1 mark]

Which description of osmosis is correct? Choose one.

A. Net movement of water from a dilute to a concentrated solution through a partially permeable membrane

B. Net movement of dissolved ions from a dilute to a concentrated solution through a partially permeable membrane

C. Net movement of water from a concentrated to a dilute solution through a partially permeable membrane

6c [1 mark]

With other conditions unchanged, which change increases diffusion rate? Choose one.

A. A larger concentration difference

B. A smaller exchange area

C. A lower temperature

7. RP02: mass, graphs and evaluation · 8 marks

7a [2 marks]

This is a separate pilot batch, not the tissue used for 7b. Use F-MASS-TABLE to calculate the percentage change in mass. Use (final mass - initial mass) ÷ initial mass × 100. Include the sign and show working.

F-MASS-TABLE · Pilot batch for 7a

F-MASS-TABLE · Pilot batch for 7a

Measurement	Recorded value
Sucrose (mol/dm³)	0.10
Initial mass / g	2.40
Final mass / g	2.64

7. RP02: mass, graphs and evaluation · continued

7b [6 marks]

The table and graph show celeriac tissue results. The investigator used one cylinder per concentration, did not blot before final weighing, and used different soaking times. They claim “every plant tissue has no net mass change at 0.40 mol/dm³”. Evaluate the claim. Use the graph and osmosis, then describe how you would improve the investigation to test the claim.

F-OSM-TABLE · Investigation for 7b

F-OSM-TABLE · Investigation for 7b

Sucrose (mol/dm³)	Mass change / %
0.00	+8
0.20	+4
0.40	0
0.60	−4
0.80	−8

One cylinder at each concentration

F-OSM-GRAPH · Investigation for 7b

